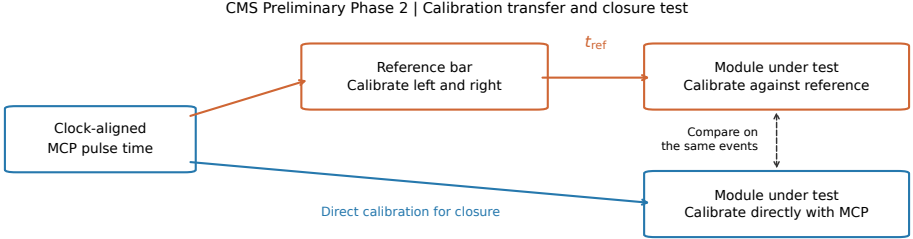
Transfer the MCP calibration to the module under test

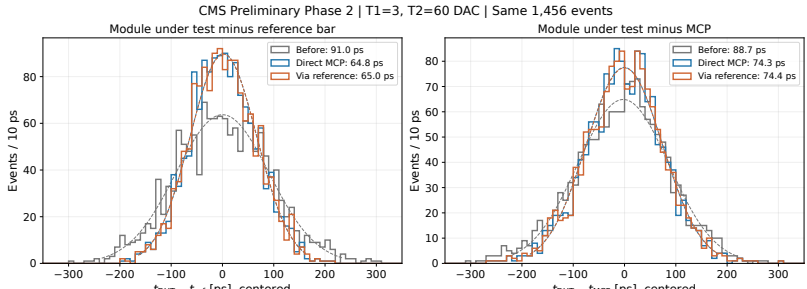
Purpose: use the calibrated reference bar when no MCP event time is available.



First form $t_{\mathrm{ref}} = \frac{1}{2}[(t_L - W_L^{\mathrm{MCP}}) + (t_R - W_R^{\mathrm{MCP}})]$. Then fit $t_j - t_{\mathrm{ref}}$ versus $1/Q_j$ for each end j of the module under test; subtract the resulting $W_j^{\mathrm{ref}}(Q_j)$.

The downstream transfer fit uses TOFHIR times and the saved reference-bar calibration, not the event's MCP time. For closure, also fit the downstream channels directly against MCP on the same selected sample. Both routes use 20 inverse-charge bins and SEM-weighted quadratics. One setting only: $T_1 = 3$, $T_2 = 60$ DAC. This tests transfer to one bar, not yet to every module or operating condition.

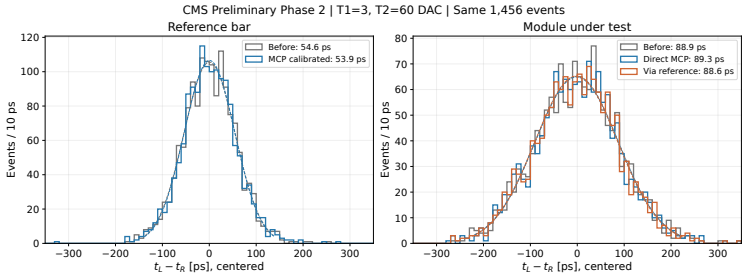
The transferred calibration agrees with direct MCP calibration



Both routes give a bar-pair width of about 65 ps, compared with 91.0 ps before correction.

Same 1456 events; common fitted charge domains, no extrapolation or threshold mixing. Bar times average the two corrected ends. Dashed curves: truncated Gaussian core fits within median $\pm 2.5(1.4826\,\mathrm{MAD})$; global median removed only for display. Width difference (via reference minus direct): $+0.24$ ps; 68% interval $[-0.30, 0.44]$ ps from 400 paired subset-bootstrap replicas. Cuts and calibrations fixed; no calibration or systematic uncertainty included. In-sample closure, not independent validation.

Left-minus-right timing checks what the 65 ps width means



The independent-end estimate $\frac{1}{2}\sqrt{\sigma^2_{L-R,\mathrm{ref}} + \sigma^2_{L-R,\mathrm{DUT}}} = 51.9\,\mathrm{ps}$ does not reproduce the 65.0 ps bar-pair width.

Same events and core-fit definition as the closure test. Left-minus-right widths change little after correction: common timing fluctuations cancel, while hit-position spread contributes. The estimate assumes independent end fluctuations and negligible position spread. The measured 65 ps is a two-bar difference width, not a single-bar resolution. Correlations and common timing terms require further study.